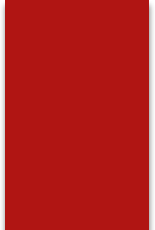
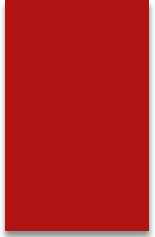


Capacitor and capacitance



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- A **capacitor** is a two-terminal electrical device that can store energy in the form of an electric charge.
 - It consists of two electrical conductors that are separated by a distance.
 - The space between the conductors may be filled by a vacuum or with an insulating material known as a dielectric.
 - The ability of the capacitor to store charges is known as **capacitance**.
 - Capacitors store energy by holding apart pairs of opposite charges. The simplest design for a capacitor is a parallel plate, which consists of two metal plates with a gap between them.



In electronic devices, energy is typically stored whether in a battery that stores energy in chemicals or in Capacitors, that store energy in an electric field.

In either case, the stored energy creates an electric potential. (One common name for that potential is voltage.) An electric potential can drive a flow of electrons. Such a flow is called an electric current. That current can be used to power electrical components within a circuit.

Engineers choose to use a battery or capacitor based on the circuit they're designing and what they want that item to do. They may even use a combination of batteries and capacitors. The devices are not totally interchangeable.



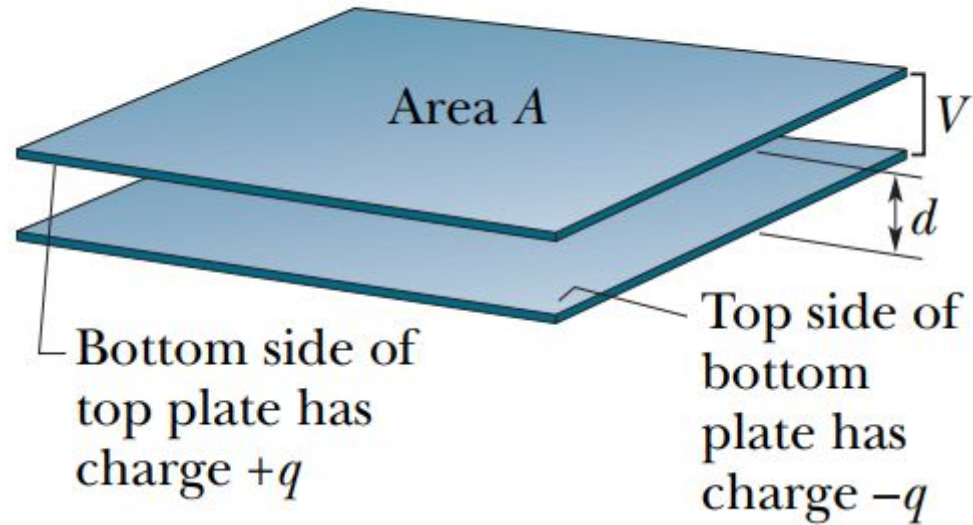
	Battery	Capacitor
Definition	Battery stores potential energy in the form of chemical energy which is later converted to the electric energy.	A Capacitor stores the potential energy in the form of electric field (electrostatic field) and release to the circuit as electric energy.
Construction	Battery has three parts known as Cathode (positive (+ve), Anode (Negative (-ve) and Separator (known as electrolyte).	Capacitor is a simple two terminal device. Terminals are metallic plates and there is a dielectric material (insulator) between them.
Function	Battery generates electrons and charge.	Capacitor only stores charged electrons.
Voltage Behavior	A battery provides almost constant voltage while discharging.	A Capacitor discharging voltage quickly decreased. i.e. discharge rate is very fast.
Charging and Discharging	The Charging and Discharging time of a Battery is slow i.e. 10 – 60+ minutes.	Charging and Discharging time of a Capacitor is very fast i.e. 1-10 seconds.
Size	For the same charging capacity, Battery size is small as compared to a capacitor	Capacitor size is large as compared to a battery for same capacity rating.

Advantages and disadvantages

A battery can store thousands of times more energy than a capacitor having the same volume. Batteries also can supply that energy in a steady, dependable stream. But sometimes they can't provide energy as quickly as it is needed.

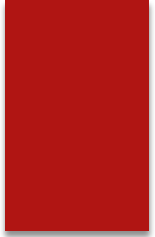
Take, for example, the flashbulb in a camera. It needs a lot of energy in a very short time to make a bright flash of light. So instead of a battery, the circuit in a flash attachment uses a capacitor to store energy. That capacitor gets its energy from batteries in a slow but steady flow. When the capacitor is fully charged, the flashbulb's "ready" light comes on. When a picture is taken, that capacitor releases its energy quickly. Then, the capacitor begins to charge up again.

Since capacitors store their energy as an electric field rather than in chemicals that undergo reactions, they can be recharged over and over again. They don't lose the capacity to hold a charge as batteries tend to do. Also, the materials used to make a simple capacitor usually aren't toxic. That means most capacitors can be tossed into the trash when the devices they power are discarded.



A parallel-plate capacitor made up of two plates of area A separated by a distance d . The charges on the facing plate surfaces have the same magnitude q but opposite signs $+q$ and $-q$

The symbol we use to represent a capacitor () is based on the structure of a parallel-plate capacitor but is used for capacitors of all geometries.



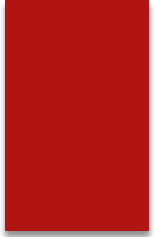
There is a potential difference between the two plates. The charge q and the potential difference V for a capacitor are proportional to each other; that is,

$$q=CV$$

The proportionality constant C is called the capacitance of the capacitor.

Its value depends only on the geometry of the plates and not on their charge or potential difference.

The **capacitance** is a measure of how much charge must be put on the plates to produce a certain potential difference between them: The greater the capacitance, the more charge is required.

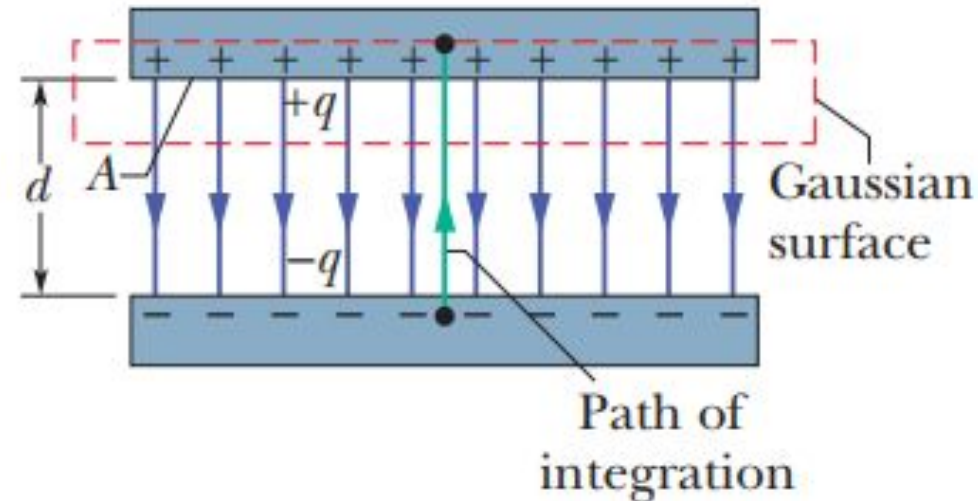


The SI **unit of capacitance** is the coulomb per volt. This unit occurs so often that it is given a special name, the **farad (F)**, $1 \text{ F} = 1 \text{ coulomb per volt}$; microfarad ($1 \mu\text{F} = 10^{-6} \text{ F}$) and the picofarad ($1 \text{ pF} = 10^{-12} \text{ F}$) are more convenient units.

On the basis of type of application, charge storing ability, and the form of the dielectric material used to separate the conductor plates, a **capacitor can be subclassified into multiple categories** such as a fixed capacitor, variable capacitor, polarized capacitor, non-polarized capacitor, electrolytic capacitor, paper capacitor, ceramic capacitor, mica capacitors, etc.

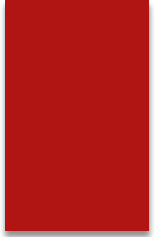
A Parallel-Plate Capacitor

We assume, as fig. suggests, that the plates of our parallel-plate capacitor are so large and so close together that we can take $\underline{E}$ to be constant throughout the region between the plates



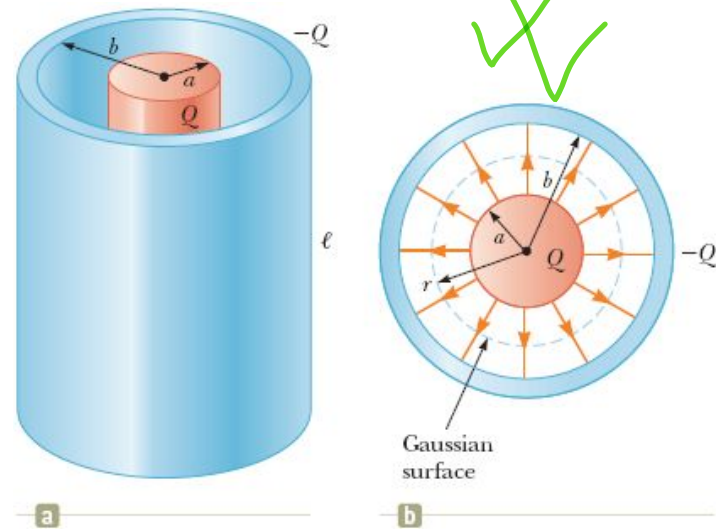
$$C = \frac{\epsilon_0 A}{d} \quad (\text{parallel-plate capacitor}).$$

Imp. note: the capacitance indeed depends only on geometrical factors—namely, the plate area A and the plate separation d . Note that C increases as we increase area A or decrease separation d

- 
1. Calculate the capacitance of an empty parallel-plate capacitor with metal plates with an area of 1.00 m^2 , separated by 1.00 mm .
 2. A parallel-plate capacitor has circular plates of 8.20 cm radius and 1.30 mm separation. (a) Calculate the capacitance. (b) Find the charge for a potential difference of 120 V .
 3. An air-filled parallel-plate capacitor has plates of area 2.30 cm^2 separated by 1.50 mm .
 - (a) Find the value of its capacitance. The capacitor is connected to a 12.0-V battery.
 - (b) What is the charge on the capacitor?
 - (c) What is the magnitude of the uniform electric field between the plates?

A Cylindrical Capacitor

Figure a shows the cylindrical capacitor including a hollow or a solid cylindrical conductor surrounded by the concentric hollow spherical cylinder, and fig. b is the cross section of a cylindrical capacitor of length l formed by two coaxial cylinders of radii a and b . We assume that $l \gg b$ to neglect fringing of electric field.



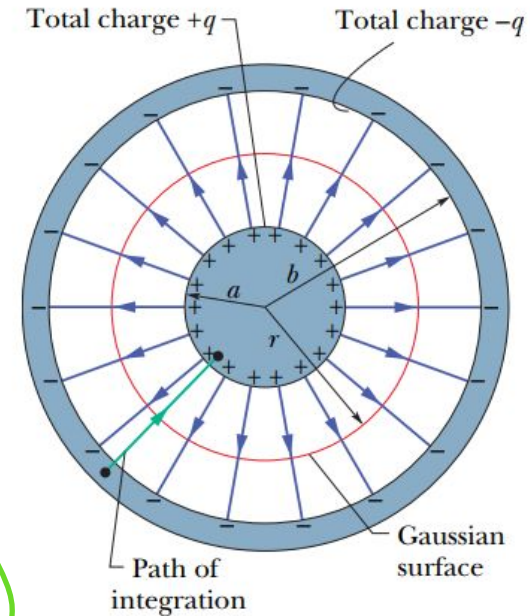
$$C = 2\pi\epsilon_0 \frac{L}{\ln(b/a)} \quad (\text{cylindrical capacitor}).$$

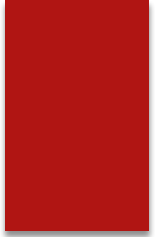
the capacitance of a cylindrical capacitor, like that of a parallel-plate capacitor, depends only on geometrical factors, in this case the length L and the two radii b and a ,

A Spherical Capacitor

Fig shows a capacitor that consists of two concentric spherical shells, of radii a and b . As a Gaussian surface, we draw a sphere of radius r concentric with the two shells.

$$C = 4\pi\epsilon_0 \frac{ab}{b - a} \quad (\text{spherical capacitor}).$$





4. The plates of a spherical capacitor have radii 38.0 mm and 40.0 mm. (a) Calculate the capacitance. (b) What must be the plate area of a parallel-plate capacitor with the same plate separation and capacitance?

5. What is the capacitance of a drop that results when two mercury spheres, each of radius $R = 2.00$ mm, merge?

6. A Cylindrical capacitor having a length of 8 cm is made of two concentric rings with an inner radius of 3 cm and an outer radius of 6 cm. Find the capacitance of the capacitor.

Energy stored in an electric field/capacitor

The electric field that builds up in the space between the plates has a direction that tends to oppose further transfer. Thus, as charge accumulates on the capacitor plates, you have to do increasingly larger amounts of work to transfer additional electrons. This work is done by a battery, at the expense of its store of chemical energy.

The work required to charge a capacitor is stored in the form of electric potential energy U in the electric field between the plates.

$$U = \frac{q^2}{2C}$$

$$U = \frac{1}{2}CV^2$$

hold no matter what the geometry of the capacitor is

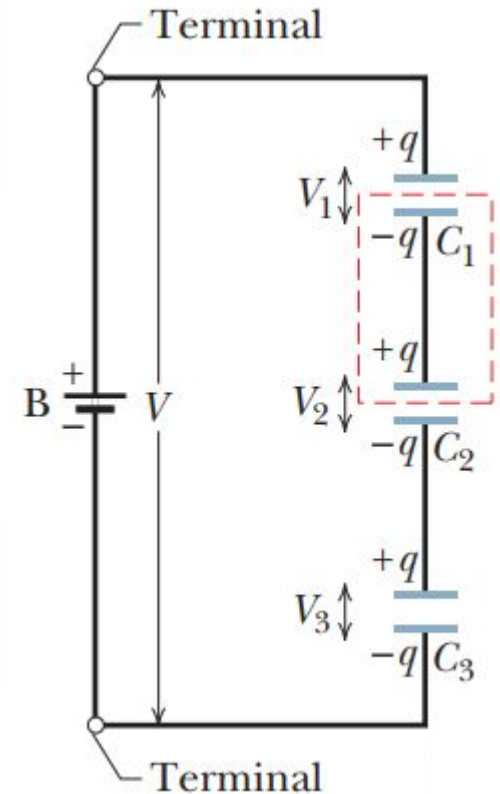
Grouping of capacitors

When there is a combination of capacitors in a circuit, we can sometimes **replace** that **combination** with an **equivalent capacitor**—that is, a single capacitor that has the same capacitance as the actual combination of capacitors.

- *With such a replacement, we can simplify the circuit, affording easier solutions for unknown quantities of the circuit.*

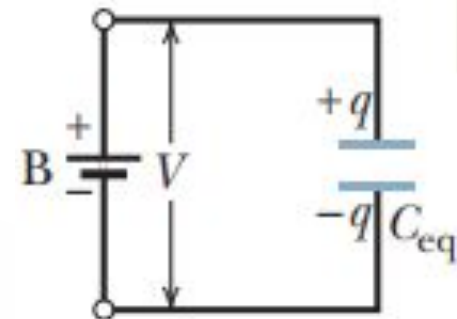
Capacitors in Series

Figure shows three capacitors connected in series to battery B, maintaining a potential difference V . The capacitors are wired serially, one after the other, and a potential difference V is applied across the two ends of the series. The potential differences that then exist across the capacitors in the series produce identical charges q on them.



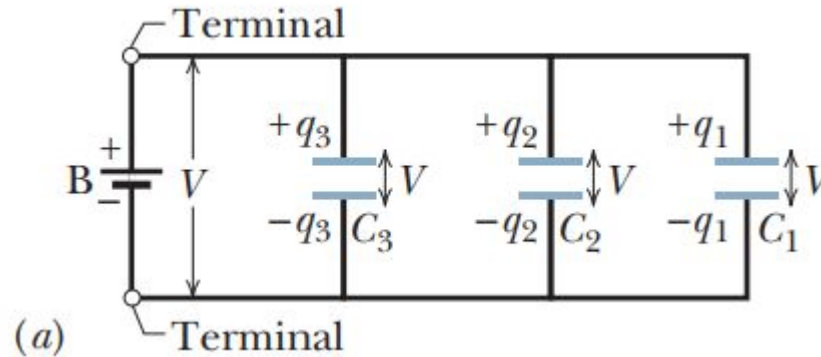
Capacitors that are connected in series can be replaced with an equivalent capacitor that has the same charge q and the same total potential difference V as the actual series capacitors.

Figure shows the equivalent capacitor (with equivalent capacitance C_{eq}) that has replaced the three actual capacitors (with actual capacitances C_1 , C_2 , and C_3)



Capacitors in Parallel

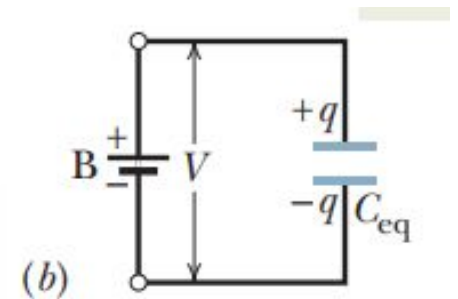
Figure a shows an electric circuit in which three capacitors are connected in parallel to battery B.



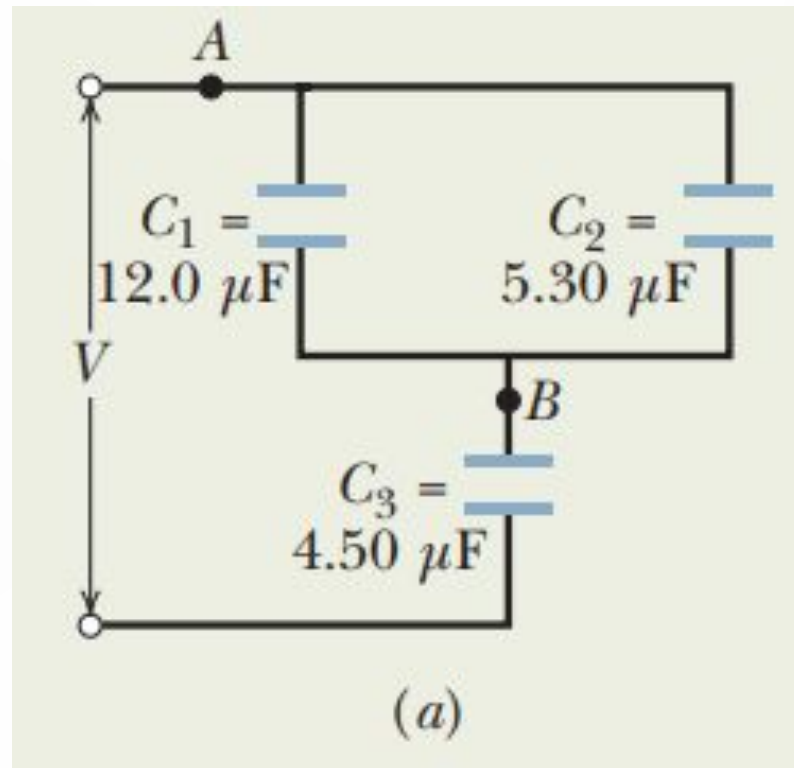
The capacitors are directly wired together at one plate and directly wired together at the other plate, and the same potential difference V is applied across the two groups of wired-together plates. Thus, each capacitor has the same potential difference V , which produces charge on the capacitor.

Capacitors connected in parallel can be replaced with an equivalent capacitor that has the same total charge q and the same potential difference V as the actual capacitors.

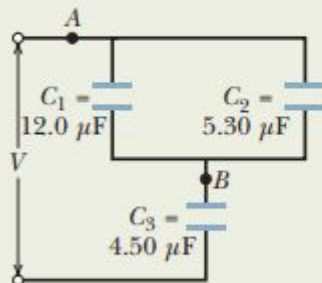
Figure b shows the equivalent capacitor (with equivalent capacitance C_{eq}) that has replaced the three capacitors (with actual capacitances C_1 , C_2 , and C_3)



1. Find (a) the equivalent capacitance for the combination of capacitances shown in Fig. a, across which potential difference V is applied.
(b) The potential difference applied to the input terminals in Fig. a is $V = 12.5 \text{ V}$. What is the charge on C_1 ?

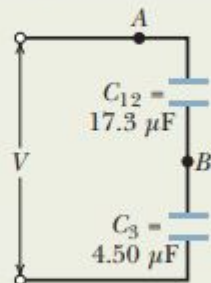


We first reduce the circuit to a single capacitor.



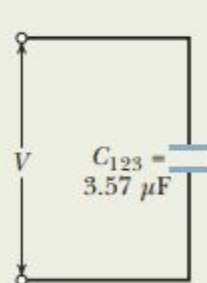
(a)

The equivalent of parallel capacitors is larger.



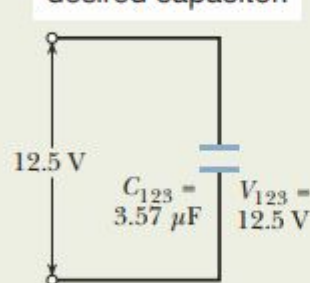
(b)

The equivalent of series capacitors is smaller.



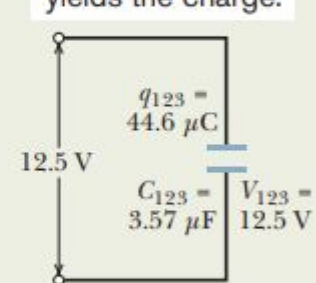
(c)

Next, we work backwards to the desired capacitor.



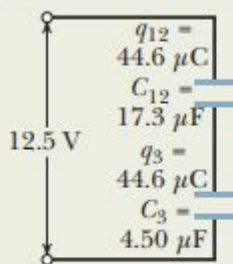
(d)

Applying $q = CV$ yields the charge.



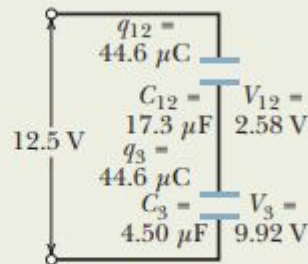
(e)

Series capacitors and their equivalent have the same q ("seri- q ").



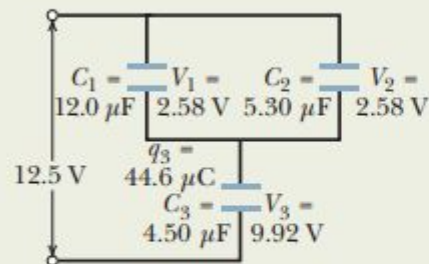
(f)

Applying $V = q/C$ yields the potential difference.



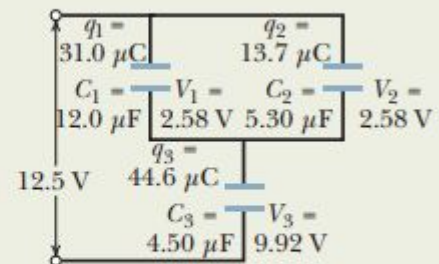
(g)

Parallel capacitors and their equivalent have the same V ("par- V ").



(h)

Applying $q = CV$ yields the charge.



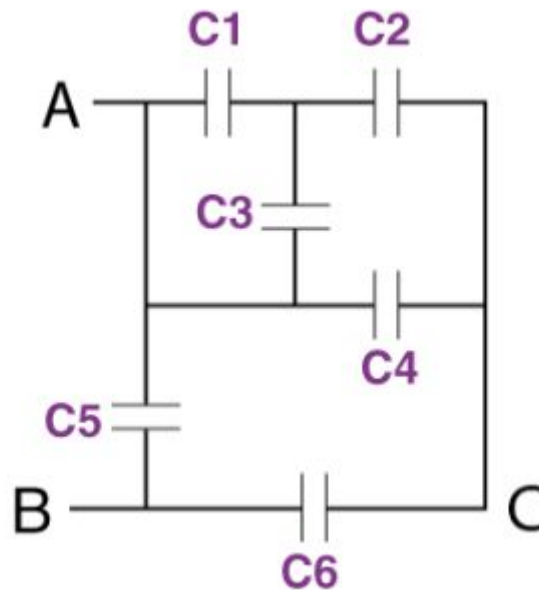
(i)

Fig. 25-10 (a) – (d) Three capacitors are reduced to one equivalent capacitor. (e) – (i) Working backwards to get the charges.

2. Two capacitors of capacitance $C_1 = 6 \mu\text{F}$ and $C_2 = 3 \mu\text{F}$ are connected in series across a cell of emf 18 V. Calculate:

- (a) The equivalent capacitance
- (b) The potential difference across each capacitor
- (c) The charge on each capacitor

3. Find the equivalent capacitance between points A and B. The capacitance of each capacitor is $2 \mu\text{F}$



Capacitor with a dielectric

If you fill the space between the plates of a capacitor with a dielectric, which is an insulating material such as mineral oil or plastic, the capacitance increases by a numerical factor k , which he called the **dielectric constant** of the insulating material.

The dielectric constant of a vacuum is unity by definition. Because air is mostly empty space, its measured dielectric constant is slightly greater than unity.

Even common paper can significantly increase the capacitance of a capacitor, and some materials, such as strontium titanate, can increase the capacitance by more than two orders of magnitude.

Some Properties of Dielectrics^a

Material	Dielectric Constant κ	Dielectric Strength (kV/mm)
Air (1 atm)	1.00054	3
Polystyrene	2.6	24
Paper	3.5	16
Transformer oil	4.5	
Pyrex	4.7	14
Ruby mica	5.4	
Porcelain	6.5	
Silicon	12	
Germanium	16	
Ethanol	25	
Water (20°C)	80.4	
Water (25°C)	78.5	
Titania ceramic	130	
Strontium titanate	310	8

For a vacuum, $\kappa = \text{unity}$.

We put insulating material between the plates of a capacitor that has been charged and then disconnected from the charging battery, as illustrated in Fig a. Because the material is insulating, the charge cannot move through it from one plate to the other, so the charge Q on the capacitor does not change.

An electric field exists between the plates of a charged capacitor, so the insulating material becomes polarized, as shown in the left figure. **An electrically insulating material that becomes polarized in an electric field is called a dielectric.**

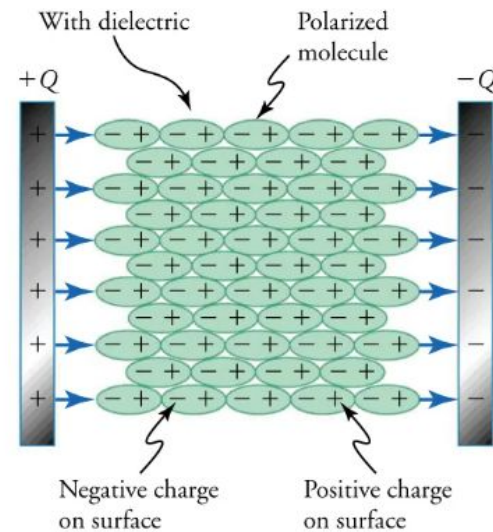
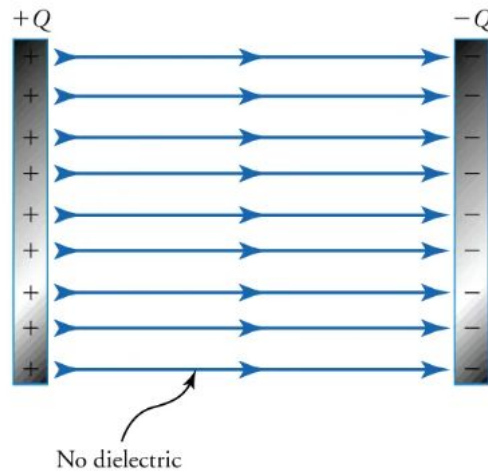
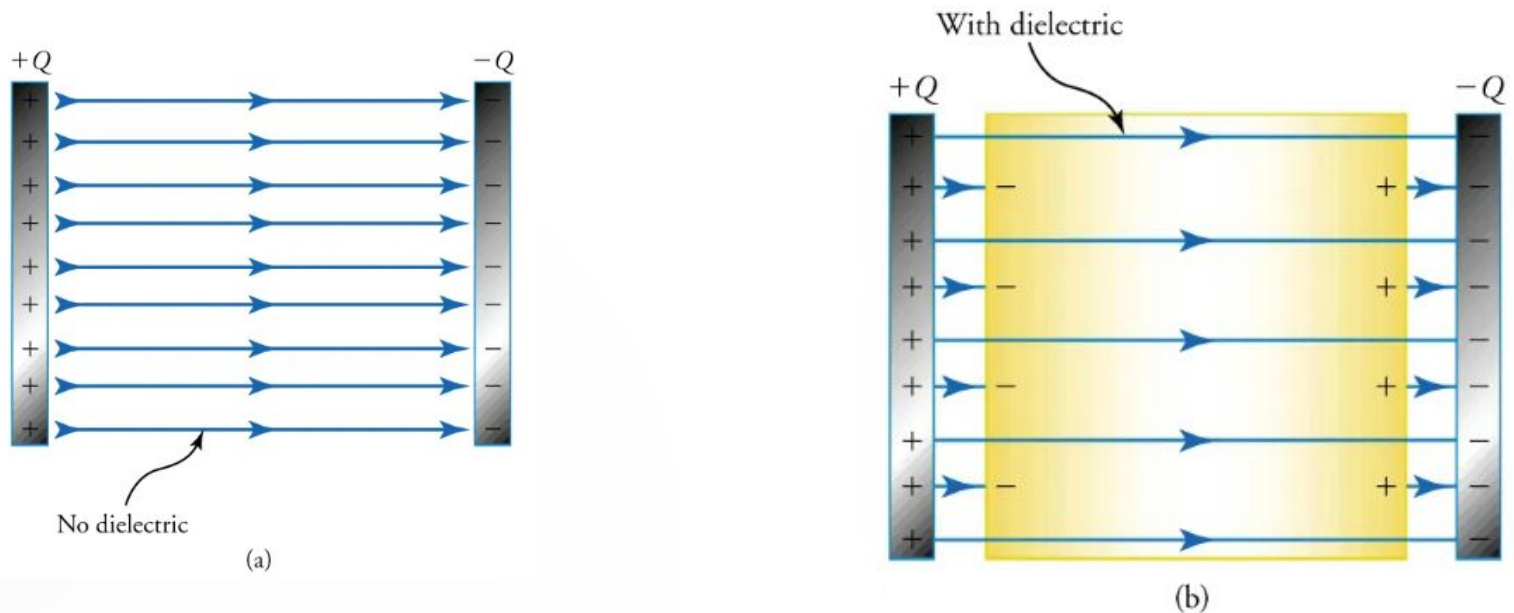


Fig b. shows that the negative charge in the molecules in the material shifts to the left, toward the positive charge of the capacitor. This shift is due to the electric field, which applies a force to the left on the electrons in the molecules of the dielectric. The right sides of the molecules are now missing a bit of negative charge, so their net charge is positive.

Fig shows a macroscopic view of a dielectric in a charged capacitor. Notice that the electric-field lines in the capacitor with the dielectric are spaced farther apart than the electric-field lines in the capacitor with no dielectric. *This means that the electric field in the dielectric is weaker than the electric field in the capacitor with no dielectric.*



The molecules in the dielectric act like tiny springs, and the energy in the electric field goes into stretching these springs. With the electric field thus weakened, the voltage difference between the two sides of the capacitor is smaller, so it becomes easier to put more charge on the capacitor. *Placing a dielectric in a capacitor before charging it therefore allows more charge and potential energy to be stored in the capacitor.*

A parallel plate with a dielectric has a capacitance of

$$C = \kappa \epsilon_0 \frac{A}{d} = \kappa C_0,$$

where κ (*kappa*) is a dimensionless constant called the *dielectric constant*. Because κ is greater than 1 for dielectrics, the capacitance increases when a dielectric is placed between the capacitor plates.

4. A typical flash for a point-and-shoot camera uses a capacitor of about $200\mu\text{F}$ (a) If the potential difference between the capacitor plates is 100 V —that is, 100 V is placed “across the capacitor,” how much energy is stored in the capacitor? (b) If the dielectric used in the capacitor were a 0.010-mm -thick sheet of nylon, what would be the surface area of the capacitor plates? dielectric constant of nylon=3.4